# TARS4 — wiring guide

The 110 TARS wires (33 in 26 AWG, the rest in 30 AWG), in build order. Each step is tested on the bench before the next; everything is done outside TARS, fitting comes at the end. The board plans (renders/ tars4_board_*.svg) give hole by hole, this guide gives both ends of every wire.

## WIRE END CODE (THE SAME ON THE BOARD PLANS)

● **Soldered**: on a board hole (plan shown) or on a module pad / lead. Tin the wire, solder, heat-shrink over the joint if it is exposed.

DF **Dupont, female contact** crimped (SN-28B, 30 AWG folded in two; two wires in one contact: together, without folding), slid into housing Xn at the cavity shown (cavity 1 = pin 1, paint dot).

DM **Dupont, male contact**: the module side of an inline pair (pigtails of X3, X7, X15).

XHF **JST-XH female** crimped, on the power block side (X10, X13, X14) or the cell side (X12).

XHM **JST-XH male** crimped, the other side of the pair.



**Wire colour**, thick = 26 AWG, strokes = black heat-shrink rings at both ends (1 = I2S / PIR, 2 = I2C, 3 = microSD).



**Power pairs**: red / black = battery and 5 V, white / blue = regulated 3V3. Rails sharing red / black carry a **coloured band** (10 mm of coloured heat-shrink, shrunk 2–3 cm from each end, the same on the + and on its GND): none = cell B+ / B−, **white** = TP4056 OUT+ / OUT−, **yellow** = VPATH, **orange** = VSYS, **purple** = VUSB. Slip the band on before soldering or crimping.

## DIAGRAM BLOCKS (FRAME COLOUR)

P **Hand-wired board**: P1, P2 (stripboards), P3 S3 carrier, P4 power path (pad perfboards); plans renders/ tars4_board_*.svg.

M **Module or part** under its name: TP4056, TPS63020, INMP441, AM312, microSD reader, switch, cells, speaker...

X **Dupont connector**, one side (female or male); cavities are numbered from pin 1.

X **JST-XH connector**, one side (female or male).

In the diagrams, TARS is seen from the front (left leg on the left), one column per TARS column, the tubes between them. A block = a board, a module or one side of a connector; each line = one wire end. A wire started at one step and finished later leaves the block with an arrow "→ st. n".

# 1 Power block, on the bench

left leg: TP4056, power-path board, TPS63020 soldered together · 22 wires finished · housings to fill: X10 F, X12 M, X13 F, X14 F, X16 M

## Before

1. Power-path board: parts, bridges and GND bus per renders/tars4_board_power.svg (Q1 / Q5 with wire-wrap wire).
2. TP4056: R_PROG replaced by 2.4 k, then tested alone: 3.7 V bench supply on B+ / B−, current < 10 µA. Polarities: C9 + on IN+, D3 band (cathode) on IN+ (reversed, it shorts the 5 V). TPS63020: EN tied to VIN, PS/SYNC to GND.
3. Crowded pads (joints ≥ 0.65 mm from the board end): splice 1 cm before the pad the two VUSB wires of IN+ (to X10 and to the power-path board), the two B+ (to X12 and X16, with the B+ sense wire to the power-path board) and the two B−: a single conductor per pad, next to C9 / D3.
4. Solder the wires below (4–5 cm loops between the three boards), then crimp the XH contacts and slide them into X10, X12, X13 and X14.
5. X12 and X16 on the block side (M): while unplugged, capped with an empty female XH housing. Their pins carry B+ / B− as soon as a cell is plugged in, until step 7.

## Bench test

- ☐ 3.7 V bench supply on X12 (block side, M): OUT+ ≈ B+; current without USB noted (D1 leakage adds to the TP4056 one: a few tens of µA possible; power_budget.md).
- ☐ X10 cavities 3–4 bridged (switch simulated): VSYS ≈ B+; 3.3 V on X13 cavity 3; VBAT/2 ≈ B+ / 2 on X14 cavity 1.
- ☐ 5 V on X10 cavities 1–2: charge LED, USB_DET ≈ 3 V on X14 cavity 2, Q1 off.

Step 1 · Power block, on the bench: diagram

## Wires finished at this step (22)

| wire | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|
| **W1** | red 26 AWG, purple band | VUSB | **P4** power path · hole F8 (wire 1 of the plan) | TP4056 · IN+ (VUSB) | — | **10** → 13 |
| **W2** | red 26 AWG, white band | OUT+ | **P4** power path · hole B4 (wire 2 of the plan) | TP4056 · OUT+ | — | **11** → 14 |
| **W3** | black 26 AWG, white band | GND | **P4** power path · hole C1 (wire 7 of the plan) | TP4056 · OUT− | — | **11** → 14 |
| **W4** | red 30 AWG | B+ | **P4** power path · hole J8 (wire 10 of the plan) | TP4056 · B+ (sense) | — | **11** → 14 |
| **W5** | red 26 AWG, orange band | VSYS | **P4** power path · hole C3 (wire 5 of the plan) | TPS63020 · VIN | — | **12** → 15 |
| **W6** | black 26 AWG, orange band | GND | **P4** power path · hole D1 (wire 8 of the plan) | TPS63020 · GND | — | **12** → 15 |
| **W7** | white 30 AWG | 3V3 | **P4** power path · hole H8 (wire 13 of the plan) | TPS63020 · VOUT | — | **12** → 15 |
| **W8** | red 26 AWG, yellow band | VPATH | **P4** power path · hole A8 (wire 3 of the plan) | XHF **X10** J1 + SW1 · case 3 VPATH | — | **11** → 14 |
| **W9** | red 26 AWG, orange band | VSYS | **P4** power path · hole D3 (wire 4 of the plan) | XHF **X10** J1 + SW1 · case 4 VSYS | — | **10** → 13 |
| **W10** | red 26 AWG, orange band | VSYS | **P4** power path · hole C3 (wire 6 of the plan) | XHF **X13** power block outputs · case 1 VSYS | — | **11** → 14 |
| **W11** | black 26 AWG, orange band | GND | **P4** power path · hole E1 (wire 9 of the plan) | XHF **X13** power block outputs · case 2 GND amp | — | **11** → 14 |
| **W12** | brown 30 AWG | VBAT/2 | **P4** power path · hole J3 (wire 11 of the plan) | XHF **X14** measurements · case 1 VBAT/2 | — | **10** → 13 |
| **W13** | orange 30 AWG | USB_DET | **P4** power path · hole E2 (wire 12 of the plan) | XHF **X14** measurements · case 2 USB_DET | — | **11** → 14 |
| **W14** | brown 30 AWG, 3 rings | SD_EN | **P4** power path · hole H5 (wire 15 of the plan) | XHF **X14** measurements · case 3 SD_EN | — | **9** → 12 |
| **W15** | red 26 AWG, purple band | VUSB | TP4056 · IN+ | XHF **X10** J1 + SW1 · case 1 VUSB | — | **9** → 12 |
| **W16** | black 26 AWG, purple band | GND | TP4056 · IN− | XHF **X10** J1 + SW1 · case 2 GND | — | **9** → 12 |
| **W17** | red 26 AWG | B+ | TP4056 · B+ | XHM **X12** left cell · case 1 B+<br>cell junction: two wires soldered on the B+ pad | — | **10** → 13 |
| **W18** | black 26 AWG | B− | TP4056 · B− | XHM **X12** left cell · case 2 B−<br>cell junction: two wires soldered on the B− pad | — | **10** → 13 |
| **W19** | red 26 AWG | B+ | TP4056 · B+ | XHM **X16** right cell · case 1 B+<br>to X16, parked above the power-path board | — | **13** → 16 |
| **W20** | black 26 AWG | B− | TP4056 · B− | XHM **X16** right cell · case 2 B− | — | **13** → 16 |
| **W21** | white 26 AWG | 3V3 | TPS63020 · VOUT | XHF **X13** power block outputs · case 3 3V3<br>along the logic GND wire | — | **11** → 14 |
| **W22** | blue 26 AWG | GND | TPS63020 · GND | XHF **X13** power block outputs · case 4 GND log<br>logic GND, along the 3V3 | — | **11** → 14 |

## Wires started here, finished later (1)

One end only now; the other stays bare until the step shown.

| wire | step | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|---|

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **W58** | **1→5** | white 30 AWG, 3 rings | SD_3V3 | **P4** power path · hole I6 (wire 14 of the plan) | DF | **X15** microSD · case 4 3V3 | — | **16** → **19** |

# 2 Charge port, switch, cells

the foot and cell cables up to X10, X12 and X16 · 6 wires finished · housings to fill: X10 M, X12 F

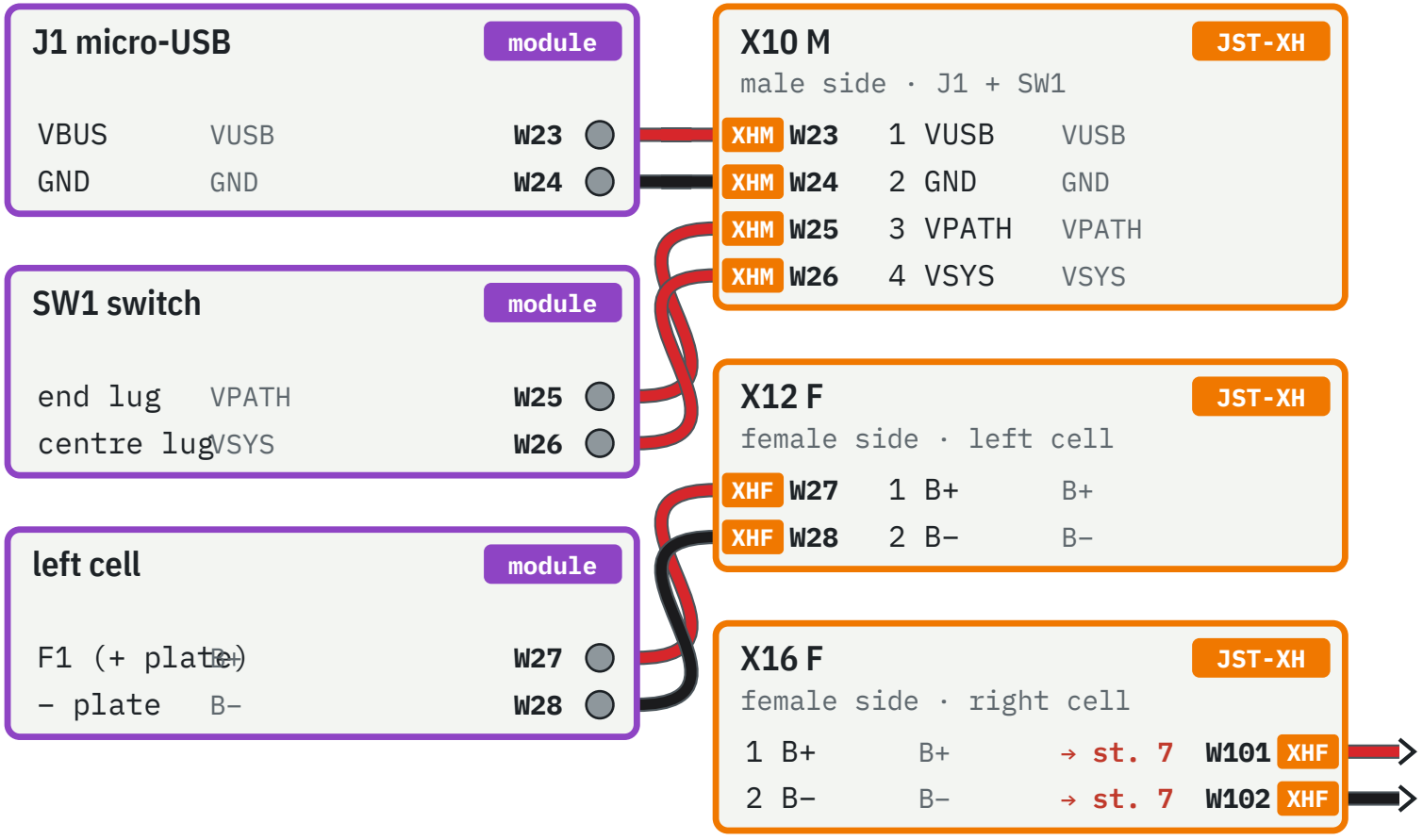
### Before

1. Solder F1 on the left + plate (F2 on the right one at step 7). Never solder on a cell.
2. Right cell wires: long enough to cross the three tubes (length from the table, already + 3 cm, offered up on TARS before cutting). Crimp X16 (female) now; the right cell end stays bare.

### Bench test

- ☐ Left cell alone: X12 plugged in, switch on, VSYS ≈ B+.
- ☐ Charger on J1: charge LED; unplugged: VUSB < 0.5 V within ~1 s.

## LEFT LEG


J1 micro-USB
module
VBUS VUSB W23
GND GND W24
SW1 switch
module
end lug VPATH W25
centre lug VSYS W26
left cell
module
F1 (+ plate) B+ W27
− plate B− W28
X10 M
JST-XH
male side · J1 + SW1
XHM W23 1 VUSB VUSB
XHM W24 2 GND GND
XHM W25 3 VPATH VPATH
XHM W26 4 VSYS VSYS
X12 F
JST-XH
female side · left cell
XHF W27 1 B+ B+
XHF W28 2 B− B−
X16 F
JST-XH
female side · right cell
1 B+ B+ → st. 7 W101 XHF
2 B− B− → st. 7 W102 XHF

## Wires finished at this step (6)

| wire | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|
| **W23** |  red 26 AWG, purple band | VUSB | ● J1 micro-USB · VBUS | XHM **X10** J1 + SW1 · case 1 VUSB<br>twisted with the GND | — | **24** → 27 |
| **W24** |  black 26 AWG, purple band | GND | ● J1 micro-USB · GND | XHM **X10** J1 + SW1 · case 2 GND | — | **24** → 27 |
| **W25** |  red 26 AWG, yellow band | VPATH | ● SW1 switch · end lug | XHM **X10** J1 + SW1 · case 3 VPATH | — | **24** → 27 |
| **W26** |  red 26 AWG, orange band | VSYS | ● SW1 switch · centre lug | XHM **X10** J1 + SW1 · case 4 VSYS | — | **24** → 27 |
| **W27** |  red 26 AWG | B+ | ● left cell · F1 (+ plate) | XHF **X12** left cell · case 1 B+<br>twisted with B– | — | **18** → 21 |
| **W28** |  black 26 AWG | B– | ● left cell · – plate | XHF **X12** left cell · case 2 B– | — | **24** → 27 |

## Wires started here, finished later (2)

One end only now; the other stays bare until the step shown.

| wire | step | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|---|
| **W101** | 2→7 |  red 26 AWG | B+ | ● right cell · F2 (+ plate) | XHF **X16** right cell · case 1 B+<br>X16 crimped at step 2, bare end threaded through the 3 tubes, soldered on F2 at step 7; twisted with B– | left tube + middle tube + right tube | **52** → 55 |
| **W102** | 2→7 |  black 26 AWG | B– | ● right cell · – plate | XHF **X16** right cell · case 2 B–<br>as B+; twisted with B+ | left tube + middle tube + right tube | **57** → 60 |

# 3 S3 carrier

the 27 controller wires, soldered on the carrier side · 0 wires finished · housings to fill: none

### Before

1. Plan renders/tars4_board_s3.svg: 2 × 8 headers (V contact removed, headers glued), standing parts, bridges, GND bus, jumpers.
2. Solder the 27 wires on the carrier side only, long enough (6–8 cm service loop + route). The other end stays bare: it is finished at the step shown.

### Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ S3 removed: each wire ↔ its pin with the ohmmeter, GND isolated from 3V3 and the signals.
- ☐ S3 plugged in, PC USB: test program flashed, the S3 boots.

Step 3 · S3 carrier: diagram

## Wires finished at this step (0)

| wire | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|

## Wires started here, finished later (27)

One end only now; the other stays bare until the step shown.

| wire | step | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|---|
| **W29** | 3→4 | white 26 AWG | 3V3 | **P3** S3 carrier · hole A4 (wire 1 of the plan) | **P1** left middle distribution · hole B9 (wire 8 of the plan) | — | **28** → 31 |
| **W30** | 3→4 | blue 26 AWG | GND | **P3** S3 carrier · hole J5 (wire 2 of the plan) | **P1** left middle distribution · hole B10 (wire 2 of the plan) | — | **30** → 33 |
| **W31** | 3→4 | red 30 AWG | DC | **P3** S3 carrier · hole A7 (wire 5 of the plan) | **P1** left middle distribution · hole A7 (wire 15 of the plan) | — | **26** → 29 |
| **W32** | 3→4 | green 30 AWG | MOSI | **P3** S3 carrier · hole C10 (wire 6 of the plan) | **P1** left middle distribution · hole A5 (wire 21 of the plan) | — | **25** → 28 |
| **W33** | 3→4 | brown 30 AWG | BL_EN | **P3** S3 carrier · hole B7 (wire 12 of the plan) | **P1** left middle distribution · hole A8 (wire 12 of the plan) | — | **27** → 30 |
| **W34** | 3→4 | orange 30 AWG | RST | **P3** S3 carrier · hole B6 (wire 13 of the plan) | **P1** left middle distribution · hole A6 (wire 18 of the plan) | — | **26** → 29 |
| **W35** | 3→4 | purple 30 AWG | CS1 | **P3** S3 carrier · hole A5 (wire 7 of the plan) | DF **X1** screen 1 · case 7 CS | — | **36** → 39 |
| **W36** | 3→4 | yellow 30 AWG | SCK1 | **P3** S3 carrier · hole C7 (wire 16 of the plan) | DF **X1** screen 1 · case 3 SCL | — | **35** → 38 |
| **W37** | 3→4 | red 30 AWG | INT1 | **P3** S3 carrier · hole A10 (wire 3 of the plan) | DF **X2** ADXL345 · case 4 INT1 | — | **22** → 25 |
| **W38** | 3→4 | green 30 AWG, 2 rings | SDA | **P3** S3 carrier · hole B9 (wire 10 of the plan) | DF **X2** ADXL345 · case 7 SDA | — | **22** → 25 |
| **W39** | 3→4 | yellow 30 AWG, 2 rings | SCL | **P3** S3 carrier · hole B8 (wire 11 of the plan) | DF **X2** ADXL345 · case 8 SCL | — | **22** → 25 |
| **W40** | 3→4 | yellow 30 AWG, 1 ring | MIC_BCLK | **P3** S3 carrier · hole B4 (wire 15 of the plan) | DF **X3** INMP441 · case 1 SCK | — | **24** → 27 |
| **W41** | 3→4 | purple 30 AWG, 1 ring | MIC_WS | **P3** S3 carrier · hole I9 (wire 18 of the plan) | DF **X3** INMP441 · case 2 WS | — | **22** → 25 |
| **W42** | 3→4 | green 30 AWG, 1 ring | MIC_SD | **P3** S3 carrier · hole I4 (wire 21 of the plan) | DF **X3** INMP441 · case 3 SD | — | **23** → 26 |
| **W59** | 3→5 | brown 30 AWG | VBAT/2 | **P3** S3 carrier · hole B11 (wire 8 of the plan) | XHM **X14** measurements · case 1 VBAT/2 | left tube | **49** → 52 |
| **W60** | 3→5 | orange 30 AWG | USB_DET | **P3** S3 carrier · hole I7 (wire 20 of the plan) | XHM **X14** measurements · case 2 USB_DET | left tube | **52** → 55 |
| **W61** | 3→5 | brown 30 AWG, 3 rings | SD_EN | **P3** S3 carrier · hole J9 (wire 22 of the plan) | XHM **X14** measurements · case 3 SD_EN | left tube | **52** → 55 |
| **W62** | 3→5 | purple 30 AWG, 3 rings | SD_CMD | **P3** S3 carrier · hole I8 (wire 19 of the plan) | DF **X15** microSD · case 2 CMD | left tube | **57** → 60 |
| **W63** | 3→5 | green 30 AWG, 3 rings | SD_D0 | **P3** S3 carrier · hole J8 (wire 23 of the plan) | DF **X15** microSD · case 3 D0 | left tube | **57** → 60 |
| **W64** | 3→5 | yellow 30 AWG, 3 rings | SD_CLK | **P3** S3 carrier · hole G11 (wire 25 of the plan) | DF **X15** microSD · case 1 CLK | left tube | **55** → 58 |
| **W73** | 3→6 | brown 30 AWG, 1 ring | AMP_SD_S3 | **P3** S3 carrier · hole A9 (wire 4 of the plan) | **P2** right middle distribution · hole A11 (wire 1 of the plan) | middle tube | **49** → 52 |
| **W74** | 3→6 | red 30 AWG, 1 ring | PIR | **P3** S3 carrier · hole B10 (wire 9 of the plan) | **P2** right middle distribution · hole A4 (wire 15 of the plan) | middle tube | **51** → 54 |
| **W75** | 3→6 | purple 30 AWG | CS2 | **P3** S3 carrier · hole B5 (wire 14 of the plan) | DF **X4** screen 2 · case 7 CS | middle tube | **47** → 50 |
| **W76** | 3→6 | yellow 30 AWG | SCK2 | **P3** S3 carrier · hole D7 (wire 17 of the plan) | DF **X4** screen 2 · case 3 SCL | middle tube | **46** → 49 |
| **W77** | 3→6 | green 30 AWG, 1 ring | DIN | **P3** S3 carrier · hole J6 (wire 24 of the plan) | DF **X5** MAX98357A · case 3 DIN | middle tube | **57** → 60 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **W78** | **3→6** | purple 30 AWG, 1 ring | LRC | **P3** S3 carrier · hole G6 (wire 26 of the plan) | **DF** | **X5** MAX98357A · case 1 LRC | middle tube | **58** → **61** |
| **W79** | **3→6** | yellow 30 AWG, 1 ring | BCLK | **P3** S3 carrier · hole G4 (wire 27 of the plan) | **DF** | **X5** MAX98357A · case 2 BCLK | middle tube | **59** → **62** |

# 4 P1 and the left middle

screen 1, mic, accelerometer · 29 wires finished · housings to fill: X1 F, X2 F, X3 F, X3 M

### Before

1. Plan renders/tars4_board_p1.svg. 1 × 8 right-angle header on the back of screen 1 (X1). ADXL345: straight header, INT2 removed and its cavity plugged (X2).
2. INMP441: 5-wire pigtail (male contacts, X3); L/R bridged to GND on its board.
3. X2 SDA and SCL: double contact, crimp together (without folding back) the S3 wire and the one leaving for P2 (bare end, soldered at step 6).

### Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ S3 on USB, switch off: screen 1 displays, the mic gives I2S samples, the ADXL345 answers at 0x53 and INT1 toggles.

Step 4 · P1 and the left middle: diagram

## Wires finished at this step (29)

| wire | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|
| **W29** | white 26 AWG | 3V3 | **P3** S3 carrier · hole A4 (wire 1 of the plan) | **P1** left middle distribution · hole B9 (wire 8 of the plan) | — | **28** → 31 |
| **W30** | blue 26 AWG | GND | **P3** S3 carrier · hole J5 (wire 2 of the plan) | **P1** left middle distribution · hole B10 (wire 2 of the plan) | — | **30** → 33 |
| **W31** | red 30 AWG | DC | **P3** S3 carrier · hole A7 (wire 5 of the plan) | **P1** left middle distribution · hole A7 (wire 15 of the plan) | — | **26** → 29 |
| **W32** | green 30 AWG | MOSI | **P3** S3 carrier · hole C10 (wire 6 of the plan) | **P1** left middle distribution · hole A5 (wire 21 of the plan) | — | **25** → 28 |
| **W33** | brown 30 AWG | BL_EN | **P3** S3 carrier · hole B7 (wire 12 of the plan) | **P1** left middle distribution · hole A8 (wire 12 of the plan) | — | **27** → 30 |
| **W34** | orange 30 AWG | RST | **P3** S3 carrier · hole B6 (wire 13 of the plan) | **P1** left middle distribution · hole A6 (wire 18 of the plan) | — | **26** → 29 |
| **W35** | purple 30 AWG | CS1 | **P3** S3 carrier · hole A5 (wire 7 of the plan) | DF **X1** screen 1 · case 7 CS | — | **36** → 39 |
| **W36** | yellow 30 AWG | SCK1 | **P3** S3 carrier · hole C7 (wire 16 of the plan) | DF **X1** screen 1 · case 3 SCL | — | **35** → 38 |
| **W37** | red 30 AWG | INT1 | **P3** S3 carrier · hole A10 (wire 3 of the plan) | DF **X2** ADXL345 · case 4 INT1 | — | **22** → 25 |
| **W38** | green 30 AWG, 2 rings | SDA | **P3** S3 carrier · hole B9 (wire 10 of the plan) | DF **X2** ADXL345 · case 7 SDA | — | **22** → 25 |
| **W39** | yellow 30 AWG, 2 rings | SCL | **P3** S3 carrier · hole B8 (wire 11 of the plan) | DF **X2** ADXL345 · case 8 SCL | — | **22** → 25 |
| **W40** | yellow 30 AWG, 1 ring | MIC_BCLK | **P3** S3 carrier · hole B4 (wire 15 of the plan) | DF **X3** INMP441 · case 1 SCK | — | **24** → 27 |
| **W41** | purple 30 AWG, 1 ring | MIC_WS | **P3** S3 carrier · hole I9 (wire 18 of the plan) | DF **X3** INMP441 · case 2 WS | — | **22** → 25 |
| **W42** | green 30 AWG, 1 ring | MIC_SD | **P3** S3 carrier · hole I4 (wire 21 of the plan) | DF **X3** INMP441 · case 3 SD | — | **23** → 26 |
| **W43** | yellow 30 AWG, 1 ring | MIC_BCLK | INMP441 · SCK | DM **X3** INMP441 · case 1 SCK<br>5–6 cm pigtail | — | **6** → 7 |
| **W44** | purple 30 AWG, 1 ring | MIC_WS | INMP441 · WS | DM **X3** INMP441 · case 2 WS | — | **6** → 7 |
| **W45** | green 30 AWG, 1 ring | MIC_SD | INMP441 · SD | DM **X3** INMP441 · case 3 SD | — | **6** → 7 |
| **W46** | white 30 AWG | 3V3 | INMP441 · VDD | DM **X3** INMP441 · case 4 VDD | — | **6** → 7 |
| **W47** | blue 30 AWG | GND | INMP441 · GND | DM **X3** INMP441 · case 5 GND<br>L/R bridged to GND on the board | — | **6** → 7 |
| **W48** | blue 30 AWG | GND | **P1** left middle distribution · hole A10 (wire 1 of the plan) | DF **X1** screen 1 · case 1 GND | — | **15** → 18 |
| **W49** | white 30 AWG | 3V3 | **P1** left middle distribution · hole A9 (wire 7 of the plan) | DF **X1** screen 1 · case 2 VCC | — | **15** → 18 |
| **W50** | brown 30 AWG | BL_EN | **P1** left middle distribution · hole B8 (wire 13 of the plan) | DF **X1** screen 1 · case 8 BLK | — | **15** → 18 |
| **W51** | red 30 AWG | DC | **P1** left middle distribution · hole B7 (wire 16 of the plan) | DF **X1** screen 1 · case 6 DC | — | **15** → 18 |
| **W52** | orange 30 AWG | RST | **P1** left middle distribution · hole B6 (wire 19 of the plan) | DF **X1** screen 1 · case 5 RES | — | **15** → 18 |
| **W53** | green 30 AWG | MOSI | **P1** left middle distribution · hole B5 (wire 22 of the plan) | DF **X1** screen 1 · case 4 SDA | — | **16** → 19 |
| **W54** | blue 30 AWG | GND | **P1** left middle distribution · hole D11 (wire 6 of the plan) | DF **X3** INMP441 · case 5 GND | — | **16** → 19 |
| **W55** | white 30 AWG | 3V3 | **P1** left middle distribution · hole D9 (wire 10 of the plan) | DF **X3** INMP441 · case 4 VDD | — | **15** → 18 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **W56** | white 30 AWG | 3V3 | DF **X3** INMP441 · case 4 VDD | DF **X2** ADXL345 · case 2 VCC<br>double contact in X3 (with the wire from P1) | – | **12** → 15 |
| **W57** | blue 30 AWG | GND | DF **X3** INMP441 · case 5 GND | DF **X2** ADXL345 · case 1 GND<br>double contact in X3 (with the wire from P1) | – | **12** → 15 |

### Wires started here, finished later (11)

One end only now; the other stays bare until the step shown.

| wire | step | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|---|
| **W65** | 4→5 | blue 26 AWG | GND | ● **P1** left middle distribution · hole C10 (wire 3 of the plan) | XHM **X13** power block outputs · case 4 GND log | left tube | **33** → 36 |
| **W66** | 4→5 | white 26 AWG | 3V3 | ● **P1** left middle distribution · hole C9 (wire 9 of the plan) | XHM **X13** power block outputs · case 3 3V3 | left tube | **33** → 36 |
| **W67** | 4→5 | blue 30 AWG, 3 rings | GND | ● **P1** left middle distribution · hole B11 (wire 4 of the plan) | DF **X15** microSD · case 5 GND | left tube | **35** → 38 |
| **W80** | 4→6 | blue 30 AWG | GND | ● **P1** left middle distribution · hole C11 (wire 5 of the plan) | ● **P2** right middle distribution · hole A10 (wire 3 of the plan) | middle tube | **29** → 32 |
| **W81** | 4→6 | white 30 AWG | 3V3 | ● **P1** left middle distribution · hole F9 (wire 11 of the plan) | ● **P2** right middle distribution · hole A9 (wire 7 of the plan) | middle tube | **29** → 32 |
| **W82** | 4→6 | brown 30 AWG | BL_EN | ● **P1** left middle distribution · hole C8 (wire 14 of the plan) | DF **X4** screen 2 · case 8 BLK | middle tube | **27** → 30 |
| **W83** | 4→6 | red 30 AWG | DC | ● **P1** left middle distribution · hole C7 (wire 17 of the plan) | DF **X4** screen 2 · case 6 DC | middle tube | **27** → 30 |
| **W84** | 4→6 | orange 30 AWG | RST | ● **P1** left middle distribution · hole C6 (wire 20 of the plan) | DF **X4** screen 2 · case 5 RES | middle tube | **27** → 30 |
| **W85** | 4→6 | green 30 AWG | MOSI | ● **P1** left middle distribution · hole C5 (wire 23 of the plan) | DF **X4** screen 2 · case 4 SDA | middle tube | **28** → 31 |
| **W86** | 4→6 | green 30 AWG, 2 rings | SDA | ● **P2** right middle distribution · hole A8 (wire 10 of the plan) | DF **X2** ADXL345 · case 7 SDA | middle tube | **36** → 39 |
| **W87** | 4→6 | yellow 30 AWG, 2 rings | SCL | ● **P2** right middle distribution · hole A7 (wire 12 of the plan) | DF **X2** ADXL345 · case 8 SCL | middle tube | **37** → 40 |

# 5 Left tube: power block ↔ left middle, microSD

everything crossing the left tube · 15 wires finished · housings to fill: X13 M, X14 M, X15 F, X15 M

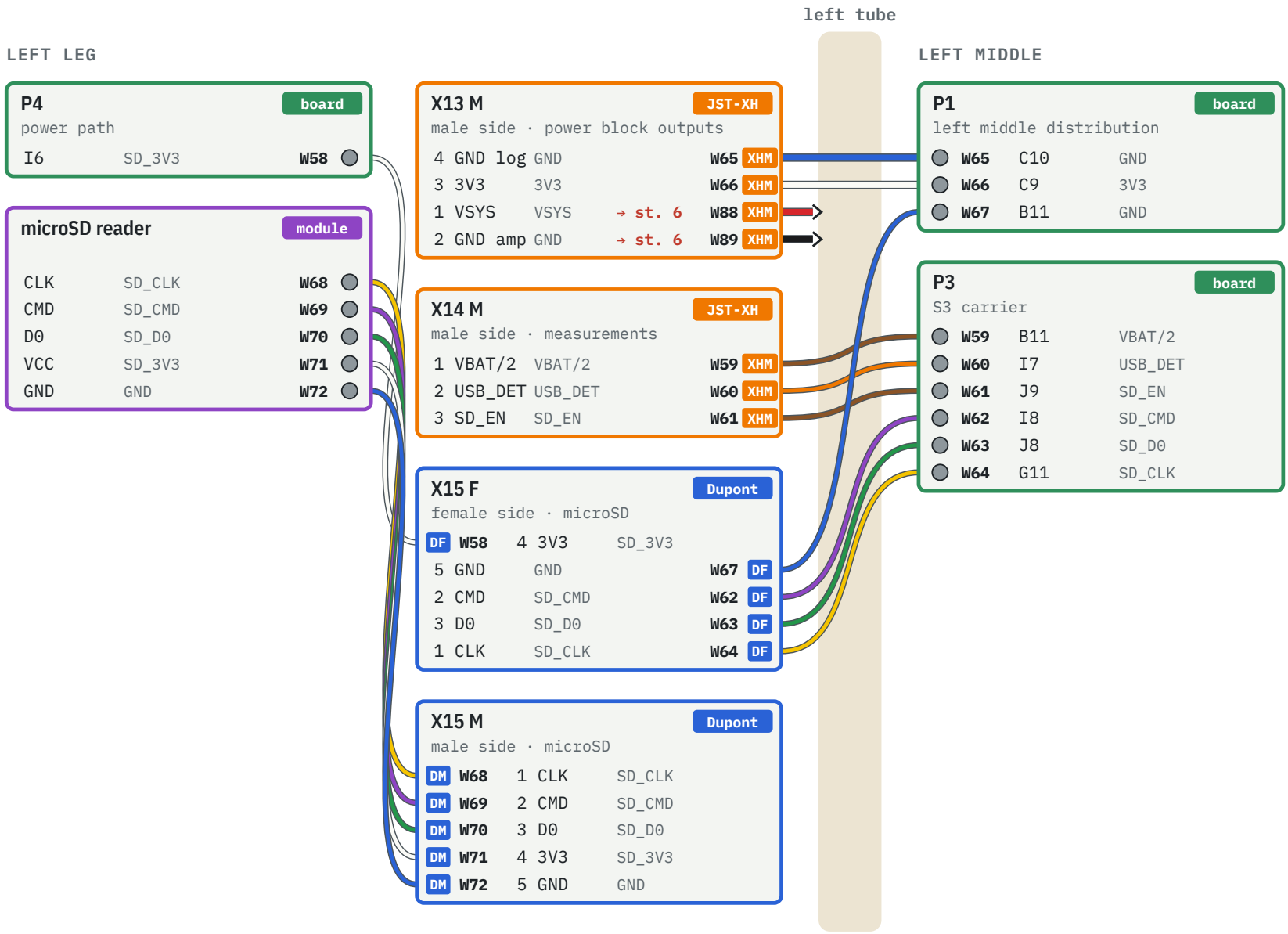
### Before

1. Thread the left brass tube onto the listed wires BEFORE crimping the far end. The amplifier pair (step 6) and the right cell pair (step 2) also cross this tube: thread them at the same time.
2. microSD reader: C4 10 µF tantalum lying flat (+ on VCC; only ~6.6 mm up to the TPS63020) and C5 100 nF on its VCC / GND pins; without pull-ups on the reader, R5–R9 10 k from CMD, D0, D1, D2 and D3 to VCC (SD_3V3). Then a 5-wire pigtail (male contacts, X15).

### Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ Without USB, switch on: the S3 boots from the cells (3.3 V on the carrier).
- ☐ VBAT/2 and USB_DET read by the S3; the microSD is read; SD_EN high cuts its 3.3 V.

Step 5 · Left tube: power block ⇔ left middle, microSD: diagram



## Wires finished at this step (15)

| wire | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|
| **W58** | white 30 AWG, 3 rings | SD_3V3 | **P4** power path · hole I6 (wire 14 of the plan) | DF **X15** microSD · case 4 3V3 | — | **16** → 19 |
| **W59** | brown 30 AWG | VBAT/2 | **P3** S3 carrier · hole B11 (wire 8 of the plan) | XHM **X14** measurements · case 1 VBAT/2 | left tube | **49** → 52 |
| **W60** | orange 30 AWG | USB_DET | **P3** S3 carrier · hole I7 (wire 20 of the plan) | XHM **X14** measurements · case 2 USB_DET | left tube | **52** → 55 |
| **W61** | brown 30 AWG, 3 rings | SD_EN | **P3** S3 carrier · hole J9 (wire 22 of the plan) | XHM **X14** measurements · case 3 SD_EN | left tube | **52** → 55 |
| **W62** | purple 30 AWG, 3 rings | SD_CMD | **P3** S3 carrier · hole I8 (wire 19 of the plan) | DF **X15** microSD · case 2 CMD | left tube | **57** → 60 |
| **W63** | green 30 AWG, 3 rings | SD_D0 | **P3** S3 carrier · hole J8 (wire 23 of the plan) | DF **X15** microSD · case 3 D0 | left tube | **57** → 60 |
| **W64** | yellow 30 AWG, 3 rings | SD_CLK | **P3** S3 carrier · hole G11 (wire 25 of the plan) | DF **X15** microSD · case 1 CLK | left tube | **55** → 58 |
| **W65** | blue 26 AWG | GND | **P1** left middle distribution · hole C10 (wire 3 of the plan) | XHM **X13** power block outputs · case 4 GND log | left tube | **33** → 36 |
| **W66** | white 26 AWG | 3V3 | **P1** left middle distribution · hole C9 (wire 9 of the plan) | XHM **X13** power block outputs · case 3 3V3 | left tube | **33** → 36 |
| **W67** | blue 30 AWG, 3 rings | GND | **P1** left middle distribution · hole B11 (wire 4 of the plan) | DF **X15** microSD · case 5 GND | left tube | **35** → 38 |
| **W68** | yellow 30 AWG, 3 rings | SD_CLK | microSD reader · CLK | DM **X15** microSD · case 1 CLK<br>5–6 cm pigtail | — | **6** → 7 |
| **W69** | purple 30 AWG, 3 rings | SD_CMD | microSD reader · CMD | DM **X15** microSD · case 2 CMD | — | **6** → 7 |
| **W70** | green 30 AWG, 3 rings | SD_D0 | microSD reader · D0 | DM **X15** microSD · case 3 D0 | — | **6** → 7 |
| **W71** | white 30 AWG, 3 rings | SD_3V3 | microSD reader · VCC | DM **X15** microSD · case 4 3V3 | — | **6** → 7 |
| **W72** | blue 30 AWG, 3 rings | GND | microSD reader · GND | DM **X15** microSD · case 5 GND | — | **6** → 7 |

## Wires started here, finished later (2)

One end only now; the other stays bare until the step shown.

| wire | step | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|---|
| **W88** | 5→6 | red 26 AWG, orange band | VSYS | XHM **X13** power block outputs · case 1 VSYS | DF **X5** MAX98357A · case 7 VIN<br>amplifier pair: threaded through the left tube at step 5, X5 end crimped here; twisted with its GND | left tube + middle tube | **48** → 51 |
| **W89** | 5→6 | black 26 AWG, orange band | GND | XHM **X13** power block outputs · case 2 GND amp | DF **X5** MAX98357A · case 6 GND<br>twisted with VSYS; the orange band tells it from the right cell B+ / B− pair in the tubes | left tube + middle tube | **48** → 51 |

# 6 Middle tube and right middle

screen 2, amplifier, speaker, AM312, P2 · 28 wires finished · housings to fill: X2 F, X4 F, X5 F, X6 F, X7 F, X7 M, X13 M

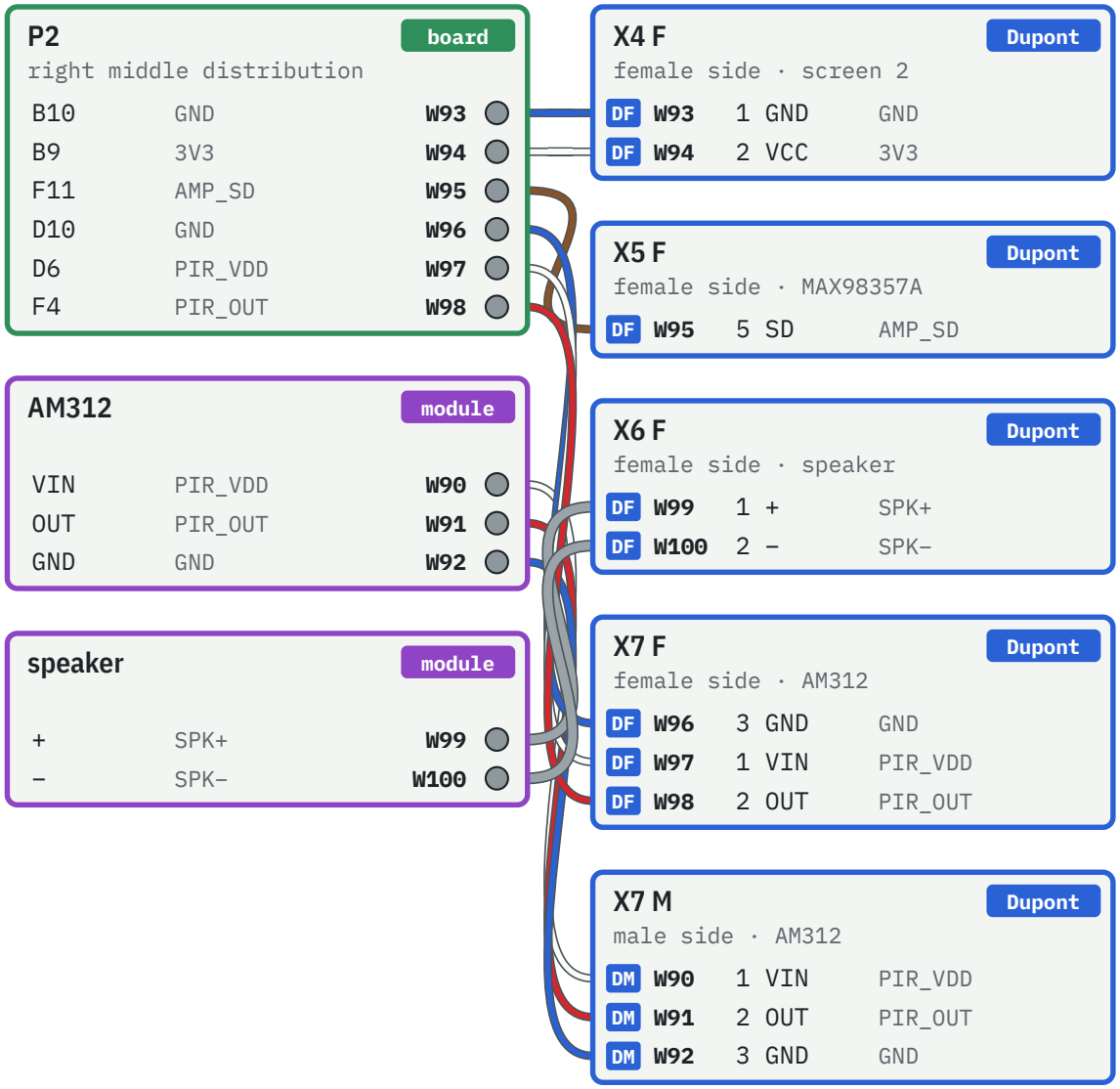
### Before

1. Plan renders/tars4_board_p2.svg. Right-angle header on the back of screen 2 (X4). Amplifier: 1 × 4 (GAIN removed) + 1 × 3 headers (X5), 2-pin header instead of the screw terminal (X6), C7 + on VIN, band (–) to GND.
2. AM312: 3-wire pigtail (male contacts, X7), breakout pins trimmed flush, breakout slipped into a heat-shrink sock (it passes 5 mm from the middle tube exit). Thread the middle tube before crimping.

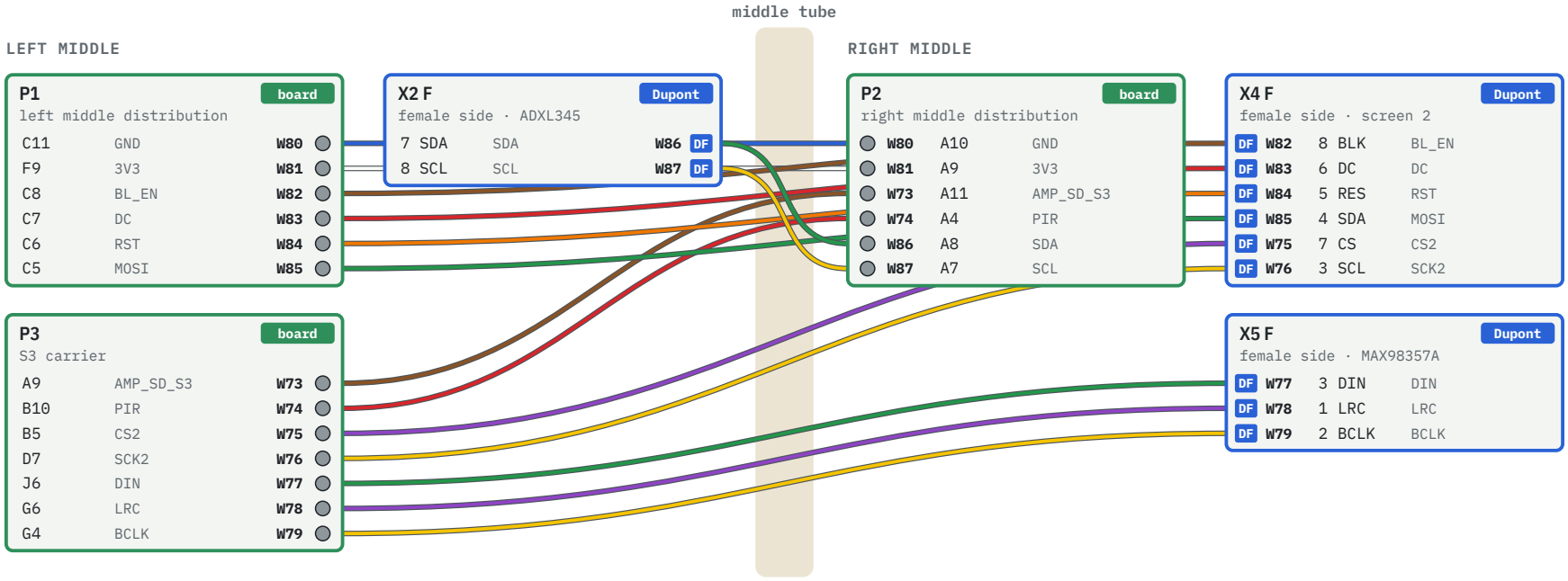
### Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ Screen 2 displays; sound in the speaker; AMP_SD low cuts the amplifier; PIR rises when a hand passes.

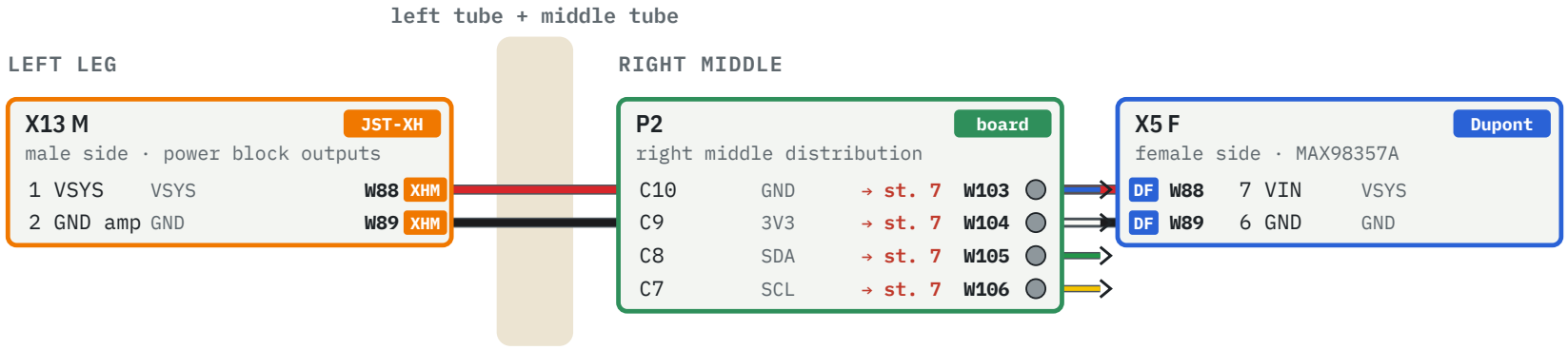
RIGHT MIDDLE



## Step 6 · Middle tube and right middle: diagram 2/3: wires through middle tube



## Step 6 · Middle tube and right middle: diagram 3/3: wires through right tube, through left tube + middle tube



## Wires finished at this step (28)

| wire | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|
| **W73** | brown 30 AWG, 1 ring | AMP_SD_S3 | **P3** S3 carrier · hole A9 (wire 4 of the plan) | **P2** right middle distribution · hole A11 (wire 1 of the plan) | middle tube | **49** → 52 |
| **W74** | red 30 AWG, 1 ring | PIR | **P3** S3 carrier · hole B10 (wire 9 of the plan) | **P2** right middle distribution · hole A4 (wire 15 of the plan) | middle tube | **51** → 54 |
| **W75** | purple 30 AWG | CS2 | **P3** S3 carrier · hole B5 (wire 14 of the plan) | DF **X4** screen 2 · case 7 CS | middle tube | **47** → 50 |
| **W76** | yellow 30 AWG | SCK2 | **P3** S3 carrier · hole D7 (wire 17 of the plan) | DF **X4** screen 2 · case 3 SCL | middle tube | **46** → 49 |
| **W77** | green 30 AWG, 1 ring | DIN | **P3** S3 carrier · hole J6 (wire 24 of the plan) | DF **X5** MAX98357A · case 3 DIN | middle tube | **57** → 60 |
| **W78** | purple 30 AWG, 1 ring | LRC | **P3** S3 carrier · hole G6 (wire 26 of the plan) | DF **X5** MAX98357A · case 1 LRC | middle tube | **58** → 61 |
| **W79** | yellow 30 AWG, 1 ring | BCLK | **P3** S3 carrier · hole G4 (wire 27 of the plan) | DF **X5** MAX98357A · case 2 BCLK | middle tube | **59** → 62 |
| **W80** | blue 30 AWG | GND | **P1** left middle distribution · hole C11 (wire 5 of the plan) | **P2** right middle distribution · hole A10 (wire 3 of the plan) | middle tube | **29** → 32 |
| **W81** | white 30 AWG | 3V3 | **P1** left middle distribution · hole F9 (wire 11 of the plan) | **P2** right middle distribution · hole A9 (wire 7 of the plan) | middle tube | **29** → 32 |
| **W82** | brown 30 AWG | BL_EN | **P1** left middle distribution · hole C8 (wire 14 of the plan) | DF **X4** screen 2 · case 8 BLK | middle tube | **27** → 30 |
| **W83** | red 30 AWG | DC | **P1** left middle distribution · hole C7 (wire 17 of the plan) | DF **X4** screen 2 · case 6 DC | middle tube | **27** → 30 |
| **W84** | orange 30 AWG | RST | **P1** left middle distribution · hole C6 (wire 20 of the plan) | DF **X4** screen 2 · case 5 RES | middle tube | **27** → 30 |
| **W85** | green 30 AWG | MOSI | **P1** left middle distribution · hole C5 (wire 23 of the plan) | DF **X4** screen 2 · case 4 SDA | middle tube | **28** → 31 |
| **W86** | green 30 AWG, 2 rings | SDA | **P2** right middle distribution · hole A8 (wire 10 of the plan) | DF **X2** ADXL345 · case 7 SDA | middle tube | **36** → 39 |
| **W87** | yellow 30 AWG, 2 rings | SCL | **P2** right middle distribution · hole A7 (wire 12 of the plan) | DF **X2** ADXL345 · case 8 SCL | middle tube | **37** → 40 |
| **W88** | red 26 AWG, orange band | VSYS | XHM **X13** power block outputs · case 1 VSYS | DF **X5** MAX98357A · case 7 VIN<br>amplifier pair: threaded through the left tube at step 5, X5 end crimped here; twisted with its GND | left tube + middle tube | **48** → 51 |
| **W89** | black 26 AWG, orange band | GND | XHM **X13** power block outputs · case 2 GND amp | DF **X5** MAX98357A · case 6 GND<br>twisted with VSYS; the orange band tells it from the right cell B+ / B− pair in the tubes | left tube + middle tube | **48** → 51 |
| **W90** | white 30 AWG | PIR_VDD | AM312 · VIN | DM **X7** AM312 · case 1 VIN<br>5–6 cm pigtail | — | **6** → 7 |
| **W91** | red 30 AWG, 1 ring | PIR_OUT | AM312 · OUT | DM **X7** AM312 · case 2 OUT | — | **6** → 7 |
| **W92** | blue 30 AWG | GND | AM312 · GND | DM **X7** AM312 · case 3 GND | — | **6** → 7 |
| **W93** | blue 30 AWG | GND | **P2** right middle distribution · hole B10 (wire 4 of the plan) | DF **X4** screen 2 · case 1 GND | — | **14** → 17 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **W94** | white 30 AWG | 3V3 | **P2** right middle distribution · hole B9 (wire 8 of the plan) | DF **X4** screen 2 · case 2 VCC | — | **15** → **18** |
| **W95** | brown 30 AWG, 1 ring | AMP_SD | **P2** right middle distribution · hole F11 (wire 2 of the plan) | DF **X5** MAX98357A · case 5 SD | — | **16** → **19** |
| **W96** | blue 30 AWG | GND | **P2** right middle distribution · hole D10 (wire 6 of the plan) | DF **X7** AM312 · case 3 GND | — | **14** → **17** |
| **W97** | white 30 AWG | PIR_VDD | **P2** right middle distribution · hole D6 (wire 14 of the plan) | DF **X7** AM312 · case 1 VIN | — | **15** → **18** |
| **W98** | red 30 AWG, 1 ring | PIR_OUT | **P2** right middle distribution · hole F4 (wire 16 of the plan) | DF **X7** AM312 · case 2 OUT | — | **17** → **20** |
| **W99** | grey 26 AWG | SPK+ | speaker · + | DF **X6** speaker · case 1 +<br>+ marked with a pen at both ends | — | **11** → **14** |
| **W100** | grey 26 AWG | SPK– | speaker · – | DF **X6** speaker · case 2 – | — | **11** → **14** |

### Wires started here, finished later (4)

One end only now; the other stays bare until the step shown.

| wire | step | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|---|
| **W103** | **6→7** | blue 30 AWG | GND | **P2** right middle distribution · hole C10 (wire 5 of the plan) | DF **X9** BME280 · case 2 GND | right tube | **27** → **30** |
| **W104** | **6→7** | white 30 AWG | 3V3 | **P2** right middle distribution · hole C9 (wire 9 of the plan) | DF **X9** BME280 · case 1 VCC | right tube | **27** → **30** |
| **W105** | **6→7** | green 30 AWG, 2 rings | SDA | **P2** right middle distribution · hole C8 (wire 11 of the plan) | DF **X9** BME280 · case 4 SDA | right tube | **27** → **30** |
| **W106** | **6→7** | yellow 30 AWG, 2 rings | SCL | **P2** right middle distribution · hole C7 (wire 13 of the plan) | DF **X9** BME280 · case 3 SCL | right tube | **28** → **31** |

# 7 Right tube and right leg

DS3231, BME280, right cell · 10 wires finished · housings to fill: X8 F, X9 F, X16 F

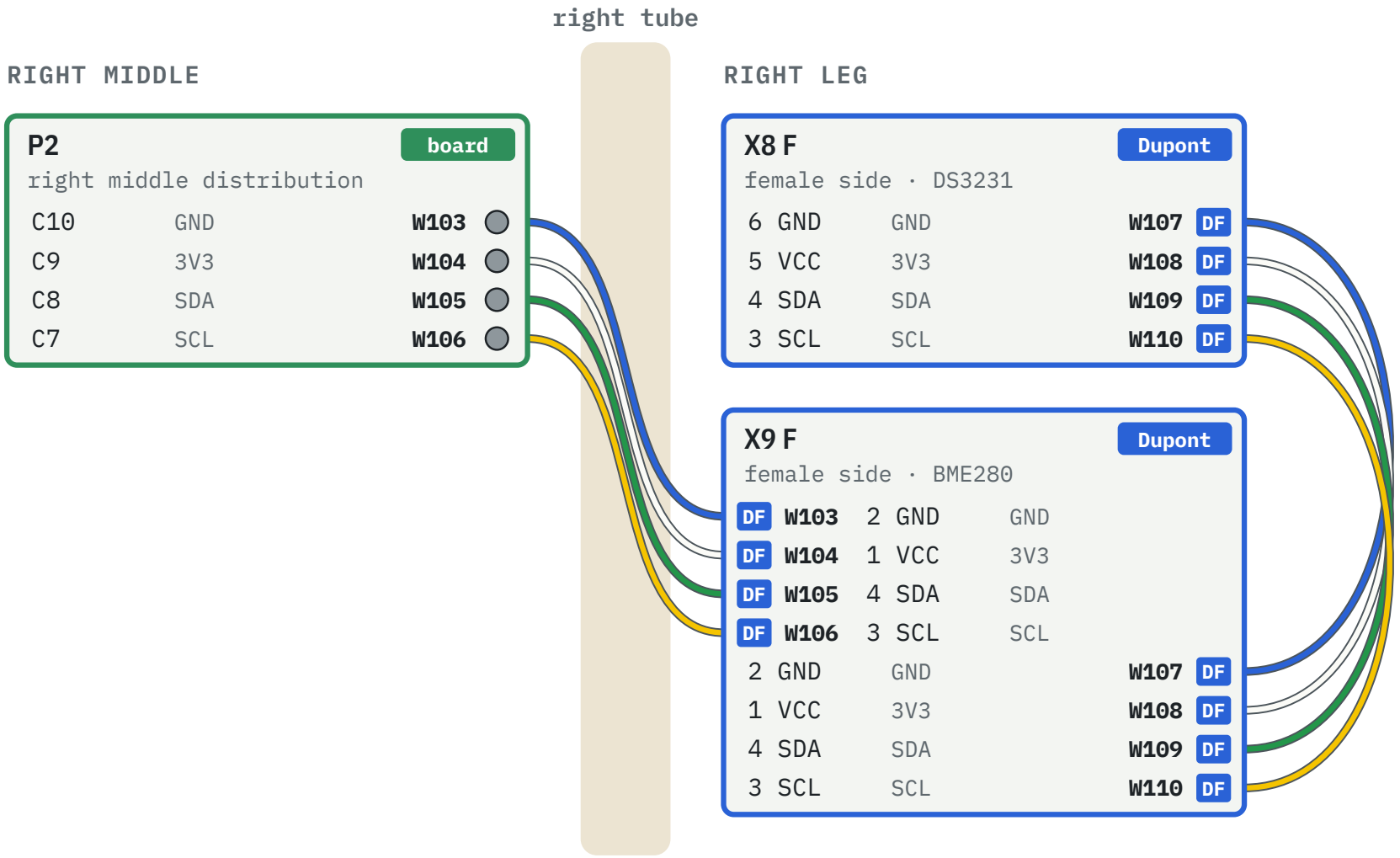
### Before

1. DS3231: right-angle header, 32K removed, coin-cell charging resistor removed. BME280: 1 × 4 header, LED removed.
2. Thread the right tube. F2 on the right + plate, then solder the two right cell wires (step 2).
3. X9: each contact takes two wires (P2 + the continuation to X8), crimped together without folding back.
4. Right tube: twist SDA with GND and SCL with 3V3 (long I2C bus, next to SCK / BCLK).

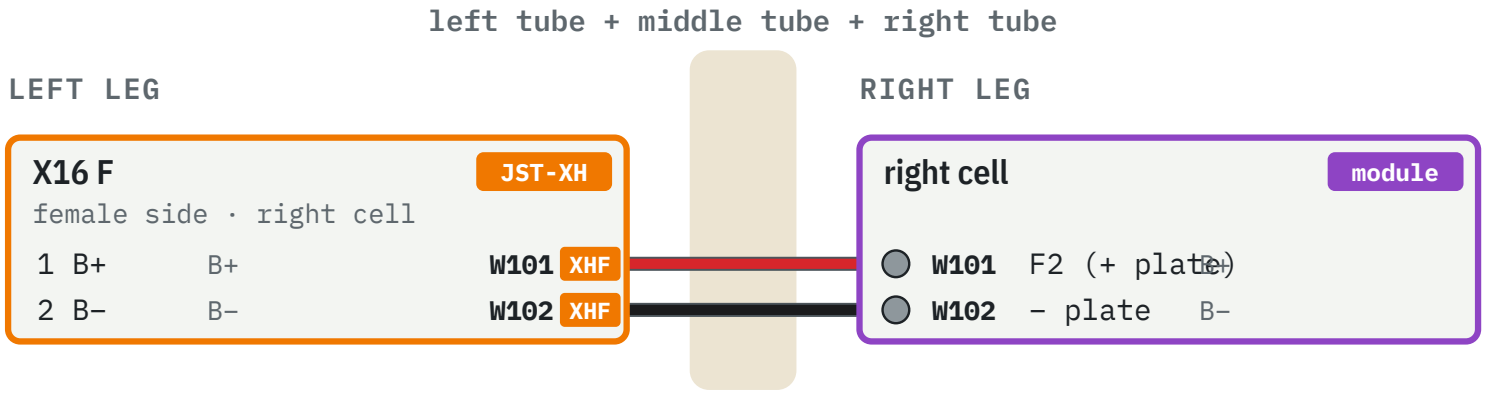
### Bench test

- ☐ I2C scan: 0x53 (ADXL345), 0x57 (DS3231 EEPROM), 0x68 (DS3231), 0x76 or 0x77 (BME280).
- ☐ Before plugging in the right cell (X16): same voltage as the left one within 0.05 V.

Step 7 · Right tube and right leg: diagram 1/2: local wires, through right tube



Step 7 · Right tube and right leg: diagram 2/2: wires through left tube + middle tube + right tube



## Wires finished at this step (10)

| wire | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|
| **W101** | red 26 AWG | B+ | right cell · F2 (+ plate) | XHF **X16** right cell · case 1 B+<br>X16 crimped at step 2, bare end threaded through the 3 tubes, soldered on F2 at step 7; twisted with B− | left tube + middle tube + right tube | **52** → 55 |
| **W102** | black 26 AWG | B− | right cell · − plate | XHF **X16** right cell · case 2 B−<br>as B+; twisted with B+ | left tube + middle tube + right tube | **57** → 60 |
| **W103** | blue 30 AWG | GND | **P2** right middle distribution · hole C10 (wire 5 of the plan) | DF **X9** BME280 · case 2 GND | right tube | **27** → 30 |
| **W104** | white 30 AWG | 3V3 | **P2** right middle distribution · hole C9 (wire 9 of the plan) | DF **X9** BME280 · case 1 VCC | right tube | **27** → 30 |
| **W105** | green 30 AWG, 2 rings | SDA | **P2** right middle distribution · hole C8 (wire 11 of the plan) | DF **X9** BME280 · case 4 SDA | right tube | **27** → 30 |
| **W106** | yellow 30 AWG, 2 rings | SCL | **P2** right middle distribution · hole C7 (wire 13 of the plan) | DF **X9** BME280 · case 3 SCL | right tube | **28** → 31 |
| **W107** | blue 30 AWG | GND | DF **X9** BME280 · case 2 GND | DF **X8** DS3231 · case 6 GND<br>double contact in X9 (with the wire from P2) | — | **14** → 17 |
| **W108** | white 30 AWG | 3V3 | DF **X9** BME280 · case 1 VCC | DF **X8** DS3231 · case 5 VCC<br>double contact in X9 (with the wire from P2) | — | **14** → 17 |
| **W109** | green 30 AWG, 2 rings | SDA | DF **X9** BME280 · case 4 SDA | DF **X8** DS3231 · case 4 SDA<br>double contact in X9 (with the wire from P2) | — | **14** → 17 |
| **W110** | yellow 30 AWG, 2 rings | SCL | DF **X9** BME280 · case 3 SCL | DF **X8** DS3231 · case 3 SCL<br>double contact in X9 (with the wire from P2) | — | **14** → 17 |

# Wire lengths

Taken from the 3D model (tars4.scad): route from one end to the other along the column, through the pivot (z 238) when the wire changes column, +10 % for bends, + service loop at each end (7 cm at the P3 carrier, 4 cm at connectors, 2–3 cm at boards and modules), + 3 cm per tube (leg rotation). Length to cut = estimated + 3 cm; module pigtails are 6 cm. Cut at assembly time, after offering the wire up in TARS if possible.

| colour | gauge | wires | to cut |
|---|---|---|---|
| yellow | 30 AWG | 11 | 3.61 m |
| green | 30 AWG | 12 | 3.49 m |
| brown | 30 AWG | 9 | 2.81 m |
| purple | 30 AWG | 7 | 2.49 m |
| blue | 30 AWG | 12 | 2.24 m |
| white | 30 AWG | 13 | 2.21 m |
| red | 30 AWG | 8 | 1.97 m |
| orange | 30 AWG | 5 | 1.46 m |
| red | 26 AWG | 15 | 3.32 m |
| black | 26 AWG | 10 | 2.49 m |
| blue | 26 AWG | 3 | 0.83 m |
| white | 26 AWG | 3 | 0.81 m |
| grey | 26 AWG | 2 | 0.28 m |
| **total 30 AWG** | | | 20.28 m |
| **total 26 AWG** | | | 7.73 m |

# Full wire list

step where it is started → finished

| wire | step | colour | signal | end A | end B | tubes | length cm<br>estimated → to cut |
|---|---|---|---|---|---|---|---|
| **W1** | 1 | red 26 AWG, purple band | VUSB | **P4** power path · hole F8 (wire 1 of the plan) | TP4056 · IN+ (VUSB) | — | **10** → **13** |
| **W2** | 1 | red 26 AWG, white band | OUT+ | **P4** power path · hole B4 (wire 2 of the plan) | TP4056 · OUT+ | — | **11** → **14** |
| **W3** | 1 | black 26 AWG, white band | GND | **P4** power path · hole C1 (wire 7 of the plan) | TP4056 · OUT− | — | **11** → **14** |
| **W4** | 1 | red 30 AWG | B+ | **P4** power path · hole J8 (wire 10 of the plan) | TP4056 · B+ (sense) | — | **11** → **14** |
| **W5** | 1 | red 26 AWG, orange band | VSYS | **P4** power path · hole C3 (wire 5 of the plan) | TPS63020 · VIN | — | **12** → **15** |
| **W6** | 1 | black 26 AWG, orange band | GND | **P4** power path · hole D1 (wire 8 of the plan) | TPS63020 · GND | — | **12** → **15** |
| **W7** | 1 | white 30 AWG | 3V3 | **P4** power path · hole H8 (wire 13 of the plan) | TPS63020 · VOUT | — | **12** → **15** |
| **W8** | 1 | red 26 AWG, yellow band | VPATH | **P4** power path · hole A8 (wire 3 of the plan) | XHF **X10** J1 + SW1 · case 3 VPATH | — | **11** → **14** |
| **W9** | 1 | red 26 AWG, orange band | VSYS | **P4** power path · hole D3 (wire 4 of the plan) | XHF **X10** J1 + SW1 · case 4 VSYS | — | **10** → **13** |
| **W10** | 1 | red 26 AWG, orange band | VSYS | **P4** power path · hole C3 (wire 6 of the plan) | XHF **X13** power block outputs · case 1 VSYS | — | **11** → **14** |
| **W11** | 1 | black 26 AWG, orange band | GND | **P4** power path · hole E1 (wire 9 of the plan) | XHF **X13** power block outputs · case 2 GND amp | — | **11** → **14** |
| **W12** | 1 | brown 30 AWG | VBAT/2 | **P4** power path · hole J3 (wire 11 of the plan) | XHF **X14** measurements · case 1 VBAT/2 | — | **10** → **13** |
| **W13** | 1 | orange 30 AWG | USB_DET | **P4** power path · hole E2 (wire 12 of the plan) | XHF **X14** measurements · case 2 USB_DET | — | **11** → **14** |
| **W14** | 1 | brown 30 AWG, 3 rings | SD_EN | **P4** power path · hole H5 (wire 15 of the plan) | XHF **X14** measurements · case 3 SD_EN | — | **9** → **12** |
| **W15** | 1 | red 26 AWG, purple band | VUSB | TP4056 · IN+ | XHF **X10** J1 + SW1 · case 1 VUSB | — | **9** → **12** |
| **W16** | 1 | black 26 AWG, purple band | GND | TP4056 · IN− | XHF **X10** J1 + SW1 · case 2 GND | — | **9** → **12** |
| **W17** | 1 | red 26 AWG | B+ | TP4056 · B+ | XHM **X12** left cell · case 1 B+<br>cell junction: two wires soldered on the B+ pad | — | **10** → **13** |
| **W18** | 1 | black 26 AWG | B− | TP4056 · B− | XHM **X12** left cell · case 2 B−<br>cell junction: two wires soldered on the B− pad | — | **10** → **13** |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **W19** | **1** | red 26 AWG | B+ | TP4056 · B+ | XHM **X16** right cell · case 1 B+<br>to X16, parked above the power-path board | — | **13** → **16** |
| **W20** | **1** | black 26 AWG | B− | TP4056 · B− | XHM **X16** right cell · case 2 B− | — | **13** → **16** |
| **W21** | **1** | white 26 AWG | 3V3 | TPS63020 · VOUT | XHF **X13** power block outputs · case 3 3V3<br>along the logic GND wire | — | **11** → **14** |
| **W22** | **1** | blue 26 AWG | GND | TPS63020 · GND | XHF **X13** power block outputs · case 4 GND log<br>logic GND, along the 3V3 | — | **11** → **14** |
| **W23** | **2** | red 26 AWG, purple band | VUSB | J1 micro-USB · VBUS | XHM **X10** J1 + SW1 · case 1 VUSB<br>twisted with the GND | — | **24** → **27** |
| **W24** | **2** | black 26 AWG, purple band | GND | J1 micro-USB · GND | XHM **X10** J1 + SW1 · case 2 GND | — | **24** → **27** |
| **W25** | **2** | red 26 AWG, yellow band | VPATH | SW1 switch · end lug | XHM **X10** J1 + SW1 · case 3 VPATH | — | **24** → **27** |
| **W26** | **2** | red 26 AWG, orange band | VSYS | SW1 switch · centre lug | XHM **X10** J1 + SW1 · case 4 VSYS | — | **24** → **27** |
| **W27** | **2** | red 26 AWG | B+ | left cell · F1 (+ plate) | XHF **X12** left cell · case 1 B+<br>twisted with B− | — | **18** → **21** |
| **W28** | **2** | black 26 AWG | B− | left cell · − plate | XHF **X12** left cell · case 2 B− | — | **24** → **27** |
| **W29** | **3→4** | white 26 AWG | 3V3 | **P3** S3 carrier · hole A4 (wire 1 of the plan) | **P1** left middle distribution · hole B9 (wire 8 of the plan) | — | **28** → **31** |
| **W30** | **3→4** | blue 26 AWG | GND | **P3** S3 carrier · hole J5 (wire 2 of the plan) | **P1** left middle distribution · hole B10 (wire 2 of the plan) | — | **30** → **33** |
| **W31** | **3→4** | red 30 AWG | DC | **P3** S3 carrier · hole A7 (wire 5 of the plan) | **P1** left middle distribution · hole A7 (wire 15 of the plan) | — | **26** → **29** |
| **W32** | **3→4** | green 30 AWG | MOSI | **P3** S3 carrier · hole C10 (wire 6 of the plan) | **P1** left middle distribution · hole A5 (wire 21 of the plan) | — | **25** → **28** |
| **W33** | **3→4** | brown 30 AWG | BL_EN | **P3** S3 carrier · hole B7 (wire 12 of the plan) | **P1** left middle distribution · hole A8 (wire 12 of the plan) | — | **27** → **30** |
| **W34** | **3→4** | orange 30 AWG | RST | **P3** S3 carrier · hole B6 (wire 13 of the plan) | **P1** left middle distribution · hole A6 (wire 18 of the plan) | — | **26** → **29** |
| **W35** | **3→4** | purple 30 AWG | CS1 | **P3** S3 carrier · hole A5 (wire 7 of the plan) | DF **X1** screen 1 · case 7 CS | — | **36** → **39** |
| **W36** | **3→4** | yellow 30 AWG | SCK1 | **P3** S3 carrier · hole C7 (wire 16 of the plan) | DF **X1** screen 1 · case 3 SCL | — | **35** → **38** |
| **W37** | **3→4** | red 30 AWG | INT1 | **P3** S3 carrier · hole A10 (wire 3 of the plan) | DF **X2** ADXL345 · case 4 INT1 | — | **22** → **25** |
| **W38** | **3→4** | green 30 AWG, 2 rings | SDA | **P3** S3 carrier · hole B9 (wire 10 of the plan) | DF **X2** ADXL345 · case 7 SDA | — | **22** → **25** |
| **W39** | **3→4** | yellow 30 AWG, 2 rings | SCL | **P3** S3 carrier · hole B8 (wire 11 of the plan) | DF **X2** ADXL345 · case 8 SCL | — | **22** → **25** |
| **W40** | **3→4** | yellow 30 AWG, 1 ring | MIC_BCLK | **P3** S3 carrier · hole B4 (wire 15 of the plan) | DF **X3** INMP441 · case 1 SCK | — | **24** → **27** |
| **W41** | **3→4** | purple 30 AWG, 1 ring | MIC_WS | **P3** S3 carrier · hole I9 (wire 18 of the plan) | DF **X3** INMP441 · case 2 WS | — | **22** → **25** |
| **W42** | **3→4** | green 30 AWG, 1 ring | MIC_SD | **P3** S3 carrier · hole I4 (wire 21 of the plan) | DF **X3** INMP441 · case 3 SD | — | **23** → **26** |
| **W43** | **4** | yellow 30 AWG, 1 ring | MIC_BCLK | INMP441 · SCK | DM **X3** INMP441 · case 1 SCK<br>5–6 cm pigtail | — | **6** → **7** |

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|---|---|---|---|---|---|---|---|
| **W44** | **4** | purple 30 AWG, 1 ring | MIC_WS | ● INMP441 · WS | DM **X3** INMP441 · case 2 WS | — | **6** → **7** |
| **W45** | **4** | green 30 AWG, 1 ring | MIC_SD | ● INMP441 · SD | DM **X3** INMP441 · case 3 SD | — | **6** → **7** |
| **W46** | **4** | white 30 AWG | 3V3 | ● INMP441 · VDD | DM **X3** INMP441 · case 4 VDD | — | **6** → **7** |
| **W47** | **4** | blue 30 AWG | GND | ● INMP441 · GND | DM **X3** INMP441 · case 5 GND<br>L/R bridged to GND on the board | — | **6** → **7** |
| **W48** | **4** | blue 30 AWG | GND | ● **P1** left middle distribution · hole A10 (wire 1 of the plan) | DF **X1** screen 1 · case 1 GND | — | **15** → **18** |
| **W49** | **4** | white 30 AWG | 3V3 | ● **P1** left middle distribution · hole A9 (wire 7 of the plan) | DF **X1** screen 1 · case 2 VCC | — | **15** → **18** |
| **W50** | **4** | brown 30 AWG | BL_EN | ● **P1** left middle distribution · hole B8 (wire 13 of the plan) | DF **X1** screen 1 · case 8 BLK | — | **15** → **18** |
| **W51** | **4** | red 30 AWG | DC | ● **P1** left middle distribution · hole B7 (wire 16 of the plan) | DF **X1** screen 1 · case 6 DC | — | **15** → **18** |
| **W52** | **4** | orange 30 AWG | RST | ● **P1** left middle distribution · hole B6 (wire 19 of the plan) | DF **X1** screen 1 · case 5 RES | — | **15** → **18** |
| **W53** | **4** | green 30 AWG | MOSI | ● **P1** left middle distribution · hole B5 (wire 22 of the plan) | DF **X1** screen 1 · case 4 SDA | — | **16** → **19** |
| **W54** | **4** | blue 30 AWG | GND | ● **P1** left middle distribution · hole D11 (wire 6 of the plan) | DF **X3** INMP441 · case 5 GND | — | **16** → **19** |
| **W55** | **4** | white 30 AWG | 3V3 | ● **P1** left middle distribution · hole D9 (wire 10 of the plan) | DF **X3** INMP441 · case 4 VDD | — | **15** → **18** |
| **W56** | **4** | white 30 AWG | 3V3 | DF **X3** INMP441 · case 4 VDD | DF **X2** ADXL345 · case 2 VCC<br>double contact in X3 (with the wire from P1) | — | **12** → **15** |
| **W57** | **4** | blue 30 AWG | GND | DF **X3** INMP441 · case 5 GND | DF **X2** ADXL345 · case 1 GND<br>double contact in X3 (with the wire from P1) | — | **12** → **15** |
| **W58** | **1→5** | white 30 AWG, 3 rings | SD_3V3 | ● **P4** power path · hole I6 (wire 14 of the plan) | DF **X15** microSD · case 4 3V3 | — | **16** → **19** |
| **W59** | **3→5** | brown 30 AWG | VBAT/2 | ● **P3** S3 carrier · hole B11 (wire 8 of the plan) | XHM **X14** measurements · case 1 VBAT/2 | left tube | **49** → **52** |
| **W60** | **3→5** | orange 30 AWG | USB_DET | ● **P3** S3 carrier · hole I7 (wire 20 of the plan) | XHM **X14** measurements · case 2 USB_DET | left tube | **52** → **55** |
| **W61** | **3→5** | brown 30 AWG, 3 rings | SD_EN | ● **P3** S3 carrier · hole J9 (wire 22 of the plan) | XHM **X14** measurements · case 3 SD_EN | left tube | **52** → **55** |
| **W62** | **3→5** | purple 30 AWG, 3 rings | SD_CMD | ● **P3** S3 carrier · hole I8 (wire 19 of the plan) | DF **X15** microSD · case 2 CMD | left tube | **57** → **60** |
| **W63** | **3→5** | green 30 AWG, 3 rings | SD_D0 | ● **P3** S3 carrier · hole J8 (wire 23 of the plan) | DF **X15** microSD · case 3 D0 | left tube | **57** → **60** |
| **W64** | **3→5** | yellow 30 AWG, 3 rings | SD_CLK | ● **P3** S3 carrier · hole G11 (wire 25 of the plan) | DF **X15** microSD · case 1 CLK | left tube | **55** → **58** |
| **W65** | **4→5** | blue 26 AWG | GND | ● **P1** left middle distribution · hole C10 (wire 3 of the plan) | XHM **X13** power block outputs · case 4 GND log | left tube | **33** → **36** |

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|---|---|---|---|---|---|---|---|
| **W66** | **4→5** | white 26 AWG | 3V3 | **P1** left middle distribution · hole C9 (wire 9 of the plan) | XHM **X13** power block outputs · case 3 3V3 | left tube | **33** → **36** |
| **W67** | **4→5** | blue 30 AWG, 3 rings | GND | **P1** left middle distribution · hole B11 (wire 4 of the plan) | DF **X15** microSD · case 5 GND | left tube | **35** → **38** |
| **W68** | **5** | yellow 30 AWG, 3 rings | SD_CLK | microSD reader · CLK | DM **X15** microSD · case 1 CLK<br>5–6 cm pigtail | — | **6** → **7** |
| **W69** | **5** | purple 30 AWG, 3 rings | SD_CMD | microSD reader · CMD | DM **X15** microSD · case 2 CMD | — | **6** → **7** |
| **W70** | **5** | green 30 AWG, 3 rings | SD_D0 | microSD reader · D0 | DM **X15** microSD · case 3 D0 | — | **6** → **7** |
| **W71** | **5** | white 30 AWG, 3 rings | SD_3V3 | microSD reader · VCC | DM **X15** microSD · case 4 3V3 | — | **6** → **7** |
| **W72** | **5** | blue 30 AWG, 3 rings | GND | microSD reader · GND | DM **X15** microSD · case 5 GND | — | **6** → **7** |
| **W73** | **3→6** | brown 30 AWG, 1 ring | AMP_SD_S3 | **P3** S3 carrier · hole A9 (wire 4 of the plan) | **P2** right middle distribution · hole A11 (wire 1 of the plan) | middle tube | **49** → **52** |
| **W74** | **3→6** | red 30 AWG, 1 ring | PIR | **P3** S3 carrier · hole B10 (wire 9 of the plan) | **P2** right middle distribution · hole A4 (wire 15 of the plan) | middle tube | **51** → **54** |
| **W75** | **3→6** | purple 30 AWG | CS2 | **P3** S3 carrier · hole B5 (wire 14 of the plan) | DF **X4** screen 2 · case 7 CS | middle tube | **47** → **50** |
| **W76** | **3→6** | yellow 30 AWG | SCK2 | **P3** S3 carrier · hole D7 (wire 17 of the plan) | DF **X4** screen 2 · case 3 SCL | middle tube | **46** → **49** |
| **W77** | **3→6** | green 30 AWG, 1 ring | DIN | **P3** S3 carrier · hole J6 (wire 24 of the plan) | DF **X5** MAX98357A · case 3 DIN | middle tube | **57** → **60** |
| **W78** | **3→6** | purple 30 AWG, 1 ring | LRC | **P3** S3 carrier · hole G6 (wire 26 of the plan) | DF **X5** MAX98357A · case 1 LRC | middle tube | **58** → **61** |
| **W79** | **3→6** | yellow 30 AWG, 1 ring | BCLK | **P3** S3 carrier · hole G4 (wire 27 of the plan) | DF **X5** MAX98357A · case 2 BCLK | middle tube | **59** → **62** |
| **W80** | **4→6** | blue 30 AWG | GND | **P1** left middle distribution · hole C11 (wire 5 of the plan) | **P2** right middle distribution · hole A10 (wire 3 of the plan) | middle tube | **29** → **32** |
| **W81** | **4→6** | white 30 AWG | 3V3 | **P1** left middle distribution · hole F9 (wire 11 of the plan) | **P2** right middle distribution · hole A9 (wire 7 of the plan) | middle tube | **29** → **32** |
| **W82** | **4→6** | brown 30 AWG | BL_EN | **P1** left middle distribution · hole C8 (wire 14 of the plan) | DF **X4** screen 2 · case 8 BLK | middle tube | **27** → **30** |
| **W83** | **4→6** | red 30 AWG | DC | **P1** left middle distribution · hole C7 (wire 17 of the plan) | DF **X4** screen 2 · case 6 DC | middle tube | **27** → **30** |
| **W84** | **4→6** | orange 30 AWG | RST | **P1** left middle distribution · hole C6 (wire 20 of the plan) | DF **X4** screen 2 · case 5 RES | middle tube | **27** → **30** |
| **W85** | **4→6** | green 30 AWG | MOSI | **P1** left middle distribution · hole C5 (wire 23 of the plan) | DF **X4** screen 2 · case 4 SDA | middle tube | **28** → **31** |
| **W86** | **4→6** | green 30 AWG, 2 rings | SDA | **P2** right middle distribution · hole A8 (wire 10 of the plan) | DF **X2** ADXL345 · case 7 SDA | middle tube | **36** → **39** |
| **W87** | **4→6** | yellow 30 AWG, 2 rings | SCL | **P2** right middle distribution · hole A7 (wire 12 of the plan) | DF **X2** ADXL345 · case 8 SCL | middle tube | **37** → **40** |

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|---|---|---|---|---|---|---|---|
| **W88** | **5→6** | red 26 AWG, orange band | VSYS | XHM **X13** power block outputs · case 1<br>VSYS | DF **X5** MAX98357A · case 7 VIN<br>amplifier pair: threaded through the left tube at step 5, X5 end crimped here; twisted with its GND | left tube + middle tube | **48** → **51** |
| **W89** | **5→6** | black 26 AWG, orange band | GND | XHM **X13** power block outputs · case 2<br>GND amp | DF **X5** MAX98357A · case 6 GND<br>twisted with VSYS; the orange band tells it from the right cell B+ / B– pair in the tubes | left tube + middle tube | **48** → **51** |
| **W90** | **6** | white 30 AWG | PIR_VDD | ● AM312 · VIN | DM **X7** AM312 · case 1 VIN<br>5–6 cm pigtail | – | **6** → **7** |
| **W91** | **6** | red 30 AWG, 1 ring | PIR_OUT | ● AM312 · OUT | DM **X7** AM312 · case 2 OUT | – | **6** → **7** |
| **W92** | **6** | blue 30 AWG | GND | ● AM312 · GND | DM **X7** AM312 · case 3 GND | – | **6** → **7** |
| **W93** | **6** | blue 30 AWG | GND | ● **P2** right middle distribution · hole B10<br>(wire 4 of the plan) | DF **X4** screen 2 · case 1 GND | – | **14** → **17** |
| **W94** | **6** | white 30 AWG | 3V3 | ● **P2** right middle distribution · hole B9<br>(wire 8 of the plan) | DF **X4** screen 2 · case 2 VCC | – | **15** → **18** |
| **W95** | **6** | brown 30 AWG, 1 ring | AMP_SD | ● **P2** right middle distribution · hole F11<br>(wire 2 of the plan) | DF **X5** MAX98357A · case 5 SD | – | **16** → **19** |
| **W96** | **6** | blue 30 AWG | GND | ● **P2** right middle distribution · hole D10<br>(wire 6 of the plan) | DF **X7** AM312 · case 3 GND | – | **14** → **17** |
| **W97** | **6** | white 30 AWG | PIR_VDD | ● **P2** right middle distribution · hole D6<br>(wire 14 of the plan) | DF **X7** AM312 · case 1 VIN | – | **15** → **18** |
| **W98** | **6** | red 30 AWG, 1 ring | PIR_OUT | ● **P2** right middle distribution · hole F4<br>(wire 16 of the plan) | DF **X7** AM312 · case 2 OUT | – | **17** → **20** |
| **W99** | **6** | grey 26 AWG | SPK+ | ● speaker · + | DF **X6** speaker · case 1 +<br>+ marked with a pen at both ends | – | **11** → **14** |
| **W100** | **6** | grey 26 AWG | SPK– | ● speaker · – | DF **X6** speaker · case 2 – | – | **11** → **14** |
| **W101** | **2→7** | red 26 AWG | B+ | ● right cell · F2 (+ plate) | XHF **X16** right cell · case 1 B+<br>X16 crimped at step 2, bare end threaded through the 3 tubes, soldered on F2 at step 7; twisted with B– | left tube + middle tube + right tube | **52** → **55** |
| **W102** | **2→7** | black 26 AWG | B– | ● right cell · – plate | XHF **X16** right cell · case 2 B–<br>as B+; twisted with B+ | left tube + middle tube + right tube | **57** → **60** |
| **W103** | **6→7** | blue 30 AWG | GND | ● **P2** right middle distribution · hole C10<br>(wire 5 of the plan) | DF **X9** BME280 · case 2 GND | right tube | **27** → **30** |
| **W104** | **6→7** | white 30 AWG | 3V3 | ● **P2** right middle distribution · hole C9<br>(wire 9 of the plan) | DF **X9** BME280 · case 1 VCC | right tube | **27** → **30** |
| **W105** | **6→7** | green 30 AWG, 2 rings | SDA | ● **P2** right middle distribution · hole C8<br>(wire 11 of the plan) | DF **X9** BME280 · case 4 SDA | right tube | **27** → **30** |
| **W106** | **6→7** | yellow 30 AWG, 2 rings | SCL | ● **P2** right middle distribution · hole C7<br>(wire 13 of the plan) | DF **X9** BME280 · case 3 SCL | right tube | **28** → **31** |
| **W107** | **7** | blue 30 AWG | GND | DF **X9** BME280 · case 2 GND | DF **X8** DS3231 · case 6 GND<br>double contact in X9 (with the wire from P2) | – | **14** → **17** |
| **W108** | **7** | white 30 AWG | 3V3 | DF **X9** BME280 · case 1 VCC | DF **X8** DS3231 · case 5 VCC<br>double contact in X9 (with the wire from P2) | – | **14** → **17** |

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|---|---|---|---|---|---|---|
| **W109** **7** |  green 30 AWG, 2 rings | SDA | DF **X9** BME280 · case 4 SDA | DF **X8** DS3231 · case 4 SDA<br>double contact in X9 (with the wire from P2) | — | **14** → **17** |
| **W110** **7** |  yellow 30 AWG, 2 rings | SCL | DF **X9** BME280 · case 3 SCL | DF **X8** DS3231 · case 3 SCL<br>double contact in X9 (with the wire from P2) | — | **14** → **17** |

## CONNECTORS (HARNESS SIDE, CAVITY 1 ON THE LEFT)

### X1 screen 1 · Dupont 1 × 8 on a right-angle header

| GND | VCC | SCL | SDA | RES |
|---|---|---|---|---|
| GND | 3V3 | SCK1 | MOSI | RST |

| DC | CS | BLK |
|---|---|---|
| DC | CS1 | BL_EN |

### X2 ADXL345 · Dupont 1 × 8, 5 wires

| GND | VCC | CS | INT1 | INT2 |
|---|---|---|---|---|
| GND | 3V3 | empty | INT1 | plugged |

| SDO | SDA | SCL |
|---|---|---|
| empty | SDA | SCL |

### X3 INMP441 · pigtail + inline Dupont 1 × 6

| SCK | WS | SD | VDD | GND |
|---|---|---|---|---|
| MIC_BCLK | MIC_WS | MIC_SD | 3V3 | GND |

| – |
|---|
| plugged |

### X4 screen 2 · Dupont 1 × 8 on a right-angle header

| GND | VCC | SCL | SDA | RES |
|---|---|---|---|---|
| GND | 3V3 | SCK2 | MOSI | RST |

| DC | CS | BLK |
|---|---|---|
| DC | CS2 | BL_EN |

### X5 MAX98357A · Dupont 1 × 4 + 1 × 3

| LRC | BCLK | DIN | GAIN | SD |
|---|---|---|---|---|
| LRC | BCLK | DIN | plugged | AMP_SD |

| GND | VIN |
|---|---|
| GND | VSYS |

### X6 speaker · Dupont 1 × 2 on a header

| + | – |
|---|---|
| SPK+ | SPK– |

### X7 AM312 · pigtail + inline Dupont 1 × 4

| VIN | OUT | GND | – |
|---|---|---|---|
| PIR_VDD | PIR_OUT | GND | plugged |

### X8 DS3231 · Dupont 1 × 6 on a right-angle header, 4 wires

| 32K | SQW | SCL | SDA | VCC |
|---|---|---|---|---|
| plugged | empty | SCL | SDA | 3V3 |

| GND |
|---|
| GND |

### X9 BME280 · Dupont 1 × 4 (daisy-chained to X8)

| VCC | GND | SCL | SDA |
|---|---|---|---|
| 3V3 | GND | SCL | SDA |

### X10 J1 + SW1 · inline JST-XH 4

| VUSB | GND | VPATH | VSYS |
|---|---|---|---|
| VUSB | GND | VPATH | VSYS |

### X12 left cell · inline JST-XH 2

| B+ | B– |
|---|---|
| B+ | B– |

### X13 power block outputs · inline JST-XH 4

| VSYS | GND amp | 3V3 | GND log |
|---|---|---|---|
| VSYS | GND | 3V3 | GND |

### X14 measurements · inline JST-XH 3

| VBAT/2 | USB_DET | SD_EN |
|---|---|---|
| VBAT/2 | USB_DET | SD_EN |

### X15 microSD · pigtail + inline Dupont 1 × 6

| CLK | CMD | D0 | 3V3 | GND |
|---|---|---|---|---|
| SD_CLK | SD_CMD | SD_D0 | SD_3V3 | GND |

| – |
|---|
| plugged |

### X16 right cell · inline JST-XH 2

| B+ | B– |
|---|---|
| B+ | B– |